

Full Length Article

Adaptive Diversity Induced Reweighting for long-tailed classification

Xiaohua Chen^a, Yucan Zhou^b, Hongcheng Li^{b,c}, Haihui Fan^b, Qinghang Su^{b,c},
Weiping Wang^{b,c}

^a Department of Automation, Tsinghua University, Beijing, China

^b Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

^c School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China

ARTICLE INFO

Keywords:

Long-tailed learning
Image classification
Reweighting
Diversity
Spanned space

ABSTRACT

Real-world large-scale data often exhibit a long-tailed distribution, making a classifier difficult to generalize well on tail categories. A straightforward and effective approach to tackling this issue is reweighting. Traditionally, the weight of a category is designed according to its *cardinality*, i.e., the number of samples it contains. Recently, some researchers claim that reweighting based on the spanned space size is more effective. They have made great progress in measuring the spanned space by analyzing sample distribution. Different from them, we aim to approximate the instinct space of a class spanned by infinite samples, which can instruct the optimizer to learn a more generalized classifier. To achieve this purpose, we maximize the space spanned by the limited training samples by sufficiently and subtly adjusting their distribution. Therefore, this paper introduces a novel metric, *diversity*, to measure the size of the maximized spanned space of a category. To quantify the class's *diversity*, we derive an easy and non-parametric formula from a new perspective of estimating its optimal prototype. Subsequently, an Adaptive Diversity Induced Reweighting (ADIR) loss is designed by allocating each class a weight inversely proportional to its *diversity*. Experiments conducted on four long-tailed datasets demonstrate that the proposed ADIR outperforms other state-of-the-art reweighting methods. Moreover, statistically balanced datasets can be imbalanced in *diversity*, where our ADIR can also achieve better results.

1. Introduction

Recently, deep neural networks have made significant breakthroughs in various tasks, where the dramatic performance is heavily reliant on well-annotated and evenly distributed datasets. However, real-world data often follow a long-tail distribution, where the majority of samples are concentrated in a few head categories, while most tail categories contain only a small number of samples (Chen, Huang, Zhu, & Lin, 2023; Deng et al., 2025; Li et al., 2024).

Obviously, head categories will dominate the training process, resulting in the separating hyperplanes heavily skewed to tail categories, particularly in large-scale long-tailed datasets. Consequently, training a classifier on a severely imbalanced dataset presents a serious challenge. An intuitive approach to achieve fair training is dataset rebalancing. This can be achieved by under-sampling head categories through random sample removal or by over-sampling tail categories through sample duplication to create a balanced training set (Byrd & Lipton, 2019; Li, Cheung, & Hu, 2023). However, under-sampling risks degrading the performance of head categories, while over-sampling

may lead to over-fitting in tail categories (Cao, Wei, Gaidon, Aréchiga, & Ma, 2019; Cui, Jia, Lin, Song, & Belongie, 2019; Wu, Huang, Liu, Wang, & Lin, 2020). Data augmentation (Chen et al., 2022; Li et al., 2021; Park, Hong, Heo, Yun, & Choi, 2022) is an effective way to diversify tail categories, but the reasonability of the augmented data warrants careful evaluation.

Compared to rebalancing, reweighting is a simpler and more light weight approach, typically involving the assignment of higher weights to tail categories in the loss functions. A typical solution (Huang, Li, Loy, & Tang, 2016; Wang, Ramanan, & Hebert, 2017) is to reweight the loss for each class in inverse proportion to the number of its training samples (i.e., *cardinality*). To achieve better results, some researchers attempt to empirically explore complex relationships between the weight and the *cardinality* (Cao et al., 2019; Zhao, Chen, Tan, Huang, & Zhu, 2022), such as relying on class sample statistics, utilizing held-out validation sets, or implementing costly training procedures (Park, Lim, Jeon, & Choi, 2021; Ren, Zeng, Yang, & Urtasun, 2018; Shu et al., 2019). Recently, several approaches (Ma, Jiao, Liu,

* Corresponding author.

E-mail addresses: chenxiaohua@mail.tsinghua.edu.cn (X. Chen), zhouyucan@iie.ac.cn (Y. Zhou), lihongcheng@iie.ac.cn (H. Li), fanhaihui@iie.ac.cn (H. Fan), suqinghang@iie.ac.cn (Q. Su), wangweiping@iie.ac.cn (W. Wang).

<https://doi.org/10.1016/j.neunet.2025.107586>

Received 18 November 2024; Received in revised form 16 April 2025; Accepted 4 May 2025

Available online 19 May 2025

0893-6080/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

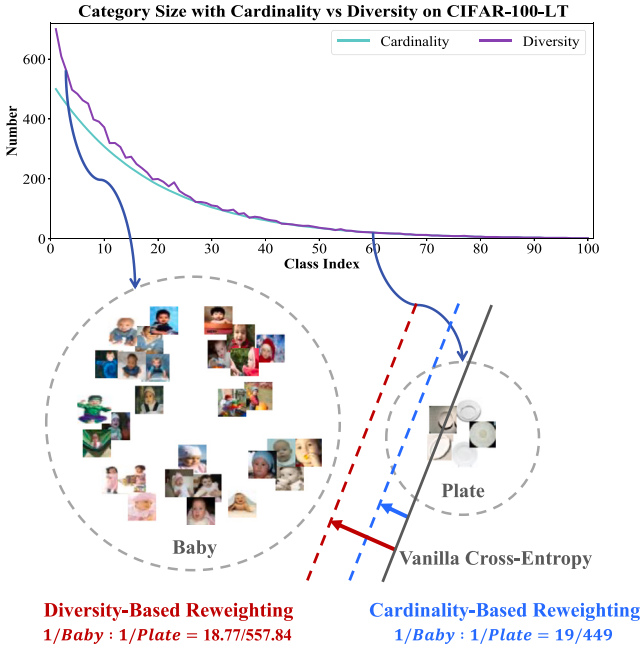


Fig. 1. The *diversity* not only considers the *cardinality*, but also relates to the distribution of samples. As samples in “Baby” are dispersed across various backgrounds, viewpoints, and poses, its *diversity* exceeds its *cardinality*. Conversely, samples in “Plate” are overlapped, so its *diversity* is smaller than the *cardinality*. When the inverse *cardinality* is used for reweighting, the separating hyperplane (blue dashed line) will still compress the space of the tail category. Therefore, the *diversity*-based reweighting can obtain a much fairer classifier (red dashed line).

Li et al., 2023; Ma, Jiao, Liu, Yang et al., 2023) have demonstrated the effectiveness of reweighting based on the size of a category’s spanned space, which is closely related to the distribution of samples. Specifically, class-balanced loss (CB) (Cui et al., 2019) is the first preliminary attempt to consider the samples’ distribution, which derives the effective number of each category with an exponential function of *cardinality*. However, it simplifies the distribution by assuming that each sample is either within the set of previous data or completely outside, which is too strict to be satisfied in the real-world scenarios. Then, CR (Ma, Jiao, Liu, Yang et al., 2023) depicts the complicated distribution of samples in a category with a perceptual manifold, and the curvature of the perceptual manifold is adopted to estimate the size of the spanned space for this category. These approaches have made great advancements in incorporating sample distribution into the reweighting process.

Different from the previous work, we consider an ideal condition, where infinite samples can be collected for a category. The optimization direction of our model is to learn the entire distribution of classes so that we can obtain a more generalized classifier. If we have enough samples, we can obtain the whole class distribution and the intrinsic space spanned by these samples. So, it is obvious that a classifier trained with the weights related to the size of intrinsic spanned space has good performance on generalization. Motivated by this, we ambitiously approximate the intrinsic spanned space with maximized space spanned by the limited training samples, which can be achieved by adjusting the weight of each sample. Once the optimal weights are derived, the category represented with the limited reweighted samples will be the most diversified. Therefore, we use *diversity*¹ to measure the size of maximized spanned space for a category. Obviously, *diversity* is also highly related to the distribution of samples. For a class, the more

similar its samples are, the smaller the *diversity* is. On the contrary, *diversity* is larger. Fig. 1 shows the size of each class depicted with *cardinality* and *diversity* on CIFAR-100-LT with the imbalance ratio of 200. Notably, the *diversity* of “Baby” 557.84 is larger than its *cardinality* 449, while the *diversity* of “Plate” 18.77 is smaller than its *cardinality* 19. Since samples of “Baby” are more diverse than those of “Plate” (with more varied backgrounds, viewpoints, and poses), it is better to give a smaller weight for “Baby” ($1/557.84 < 1/449$) and a larger weight for “Plate” ($1/18.77 > 1/19$). Therefore, the *diversity*-based reweighting will help to achieve fairer training.

In this paper, we propose to achieve fairer separating hyperplanes with the *diversity*-based reweighting. We define and quantify the *diversity* of a category from the perspective of estimating its optimal prototype, which implies the maximum *diversity* of all its samples. During the estimation of the optimal prototype, we derive a simple and non-parametric formula $\sum_{i,j}(\mathbb{P}^{-1})_{ij}$ to calculate the *diversity* of a category, where $\mathbb{P}$ is the correlation matrix and each element denotes the correlation coefficient between two samples in this category. Then, an Adaptive Diversity Induced Reweighting (ADIR) is proposed to alleviate the challenges of long-tailed classification. Specifically, the training process consists of two stages. In stage-I, we train a basic classification model using the cross-entropy loss. In stage-II, we calculate each $\mathbb{P}_{ij}$ with the extracted features. Then, *diversity* of each category can be calculated by $\sum_{i,j}(\mathbb{P}^{-1})_{ij}$. Subsequently, the loss is reweighted by inverting the *diversity* to achieve *diversity*-balanced learning. It is notable that weights in our ADIR are adaptively updated because they are calculated with $\mathbb{P}_{ij}$, which is highly related to the optimizing features.

Our key contributions are:

- We use *diversity* to measure the size of the maximized spanned space for a category and investigate its intrinsic advantages. By assigning the inverse of *diversity* as the weights for each class, the simply improved loss can achieve fairer separating hyperplanes.
- Based on the correlation between all the samples in a category, we derive a simple and non-parametric formula to calculate its *diversity* by estimating the optimal prototype.
- Extensive experimental results show that our ADIR outperforms current state-of-the-art (SOTA) reweighting methods. Furthermore, it also obtains good improvements on balanced CIFAR.

In the rest of our content, we will introduce the structure of the article from the following aspects. Section 2 introduces the work related to this method, including approaches such as Rebalancing, Reweighting, SOTA with “whistles and bells”. Section 3 presents an analysis of the traditional prototype calculation way, while Section 4 derives the formulation to estimate the *diversity* of a class, which is subsequently used for reweighting. Our method consists of two stages, this Section also describes the implementation of ADIR algorithm. The experimental results across different datasets are presented in Section 5. Finally, we describe the conclusions and outline future work in Section 6.

2. Related work

We review the two most classic types of methods in long-tailed classification, including Rebalancing and Reweighting. Additionally, we review the state-of-the-art (SOTA) ‘whistles and bells’ methods in long-tailed classification.

Rebalancing. A widely used rebalancing strategy to address the long-tailed problem is re-sampling, including under-sampling and over-sampling (Zhang, Kang, Hooi, Yan, & Feng, 2021). Over-sampling randomly duplicates samples from tail class, while under-sampling randomly discards samples in head class, so as to achieve a balanced training set. Decoupling (Kang et al., 2020) empirically finds that instance-balanced sampling is beneficial for representation learning. Meanwhile, BBN (Zhou, Cui, Wei, & Chen, 2020) designs two branches

¹ Italic *diversity* represents the size of maximized spanned space for a category.

combining instance-balanced and class-balanced sampling to achieve better results. Balanced meta-softmax (Ren et al., 2020) learns the optimal sampling probability for each class by employing the classification loss with a validation set. Although the above re-sampling methods have improved the performance, it may easily cause overfitting on the tail.

To alleviate overfitting, data augmentation (Chou, Chang, Pan, Wei, & Juan, 2020; Kim, Jeong, & Shin, 2020) is introduced to enrich the diversity. An intuitive approach to generating new instances is combining similar samples from a target class (Chu, Bian, Liu, & Ling, 2020). Mixup (Zhang, Cissé, Dauphin, & Lopez-Paz, 2018) creates new samples by a convex combination of two randomly selected images from the target class. Recently, RSG (Wang, Lukasiewicz, Hu, Cai, & Xu, 2021) enhances the diversity for tail class by transferring the feature displacement from head to tail while CMO (Park et al., 2022) leverages the rich context of the head as background images to diversify the tail samples. In addition to the above explicit augmentation methods, MetaSAug (Li, Kaixiong et al., 2021) pursues implicit augmentation. It attempts to use the class-wise covariance matrix with rich semantic transformations to obtain diverse samples. However, the diversity of the tail covariance matrix is limited since the tail class only has a few samples. Therefore, RISDA (Chen et al., 2022) proposes to transfer transformations from all related classes with a knowledge graph to get an optimal diversified covariance matrix. Existing augmentation methods have achieved great progress.

Reweighting. Another typical strategy involves reweighting the loss to emphasize the tail classes, including both instance-level and class-level reweighting methods. Instance-level reweighting allocate distinct weights to each instance with the assumption that samples act differently in optimizing the model. Focal loss (Lin, Goyal, Girshick, He, & Dollár, 2017) assigns higher weights to difficult samples based on the prediction probability. L2RW (Ren et al., 2018) and Meta-Weight-Net (Shu et al., 2019) introduce an online reweighting approach that leverages an extra validation set to dynamically adjust sample weights. However, it is expensive to obtain a held-out validation set in the long-tailed scenario. Therefore, FSR (Zhang & Pfister, 2021) uses an unbiased dictionary to provide reward signals to adjust the weights for each sample. Recently, IB (Park et al., 2021) down-weights the hard samples with great influence on the hyperplane to alleviate the effect of overfitting. Although the instance-level method is effective, the elaborate weight assignment is complex in implementation.

Differently, class-level reweighting is simpler and lightweight, which assigns larger weights for the tail categories. An intuitive approach is to reweight each class inversely proportional to its *cardinality* (Wang, Chen, & Hoi, 2021). However, this strategy performs worse on real-world, long-tailed datasets. Therefore, some methods empirically explore complex relationships between weight and *cardinality* (Li, Cheung, & Lu, 2022). Meta-class-weight (Jamal, Brown, Yang, Wang, & Gong, 2020) improves weight estimation by leveraging meta-learning. LDAM (Cao et al., 2019) introduces a marginal loss based on *cardinality* to extend the hyperplane for the tail. Whereas, the spanned space of a class extends far beyond its *cardinality*, as the distribution of samples is also significant. Several researchers have shown that reweighting based on the spanned space size by considering the samples' distribution can achieve fairer training. Specifically, the class-balanced loss is the first attempt to emphasize the effective number of a class by considering the overlap among its samples. However, the strong assumption that a sample is either entirely within or entirely outside the set of previous data is challenging to satisfy in real-world scenarios, as samples from the same class often exhibit partial overlap. Additionally, the effective number is denoted based on an exponential function of *cardinality*, yet it still neglects the complex distribution of samples within the feature space. Therefore, CR (Ma, Jiao, Liu, Yang et al., 2023) proposes using the curvature of a perceptual manifold to sufficiently consider the sample distribution in a category. This distinct metric effectively measures the spanned size of a category and has achieved excellent reweighting by

incorporating sample distribution, capturing its complexity. However, it estimates the class size based on the available training samples, whose distribution is biased toward dominant patterns (e.g., front views, clear backgrounds, etc.), leading to potentially skewed metrics.

Different from their work, we consider an ideal distribution of samples. When there are infinite samples can be collected for a class, the space spanned by them is the intrinsic space of it. Obviously, weights related to the size of intrinsic space for reweighting can achieve a more generalized classifier. To approximate the intrinsic class size, we quantify the size of the maximized space spanned with the current training set, which can be achieved by adjusting the weight of each sample. With the adjusted distribution of samples, we propose a new metric, *diversity*, to represent the size of maximized spanned space for a class. Then, simply taking the inverse of *diversity* as the class weight can significantly enhance the performance of long-tailed classification.

SOTA with “whistles and bells”. Long-tailed classification is a typical issue with many works recently (Sinha, Ohashi, & Nakamura, 2022; Wu, Min, Zhang, & Wang, 2023). In addition to the classical rebalancing and reweighting methods, numerous state-of-the-art (SOTA) approaches have emerged in recent years. These methods typically include ensemble-based methods that integrate multiple models, contrastive learning-based approaches, and pretrained-based methods.

Among these, ensemble-based methods address the long-tailed visual learning problem by strategically combining multiple network modules (i.e., multiple experts). These methods follow a complementary learning paradigm, where different experts specialize in distinct aspects of the task, with each expert benefiting from its specialization in the dominant component. Final decisions are made through joint inference based on the predictions of each expert. LFME (Xiang, Ding, & Han, 2020) is inspired by the idea that learning from a more evenly distributed set of samples is sometimes easier than learning from a long-tailed distribution with all samples. To promote expert diversity, LFME achieves this by learning from different dataset groupings and then applying knowledge distillation from multiple teacher experts to train a unified student model. NCL (Li, Tan, Wan, Lei, & Guo, 2022) introduces a nested collaborative learning approach for multi-expert networks, facilitating the learning of both a comprehensive perspective across all categories and a focused perspective on challenging categories. This enables each expert to gain supplementary knowledge from the other experts. TLC (Li, Han, Li, Fu, & Zhang, 2022) presents a multi-expert fusion strategy that leverages the uncertainty of each expert. This approach enhances classification performance and reliability by effectively identifying and addressing hard samples.

Recently, some studies have found that the performance of supervised contrastive loss outperforms traditional cross-entropy loss in image recognition tasks (Khosla et al., 2020). Inspired by this, some long-tail classification methods based on supervised contrastive learning (SCL) have emerged to improve representation learning for long-tail distributions. Hybrid (Wang, Han, Wei, Zhang, & Wang, 2021) proposes a novel hybrid network architecture that enhances model performance by designing a feature representation learning branch based on supervised prototype contrastive learning and a classifier branch based on cross-entropy loss. KCL (Kang, Li, Xie, Yuan, & Feng, 2021) introduces a contrastive loss with K-positive pairs to balance the feature space. By ensuring that all classes in the KCL positive pair construction use the same number of samples, this approach helps alleviate class imbalance and improves the model's generalization ability. TSC (Li et al., 2022) achieves a balanced feature space by forcing features from different classes to converge to a set of corresponding target points with a uniform distribution during training.

With the rise of vision-language pretraining, which leverages contrastive learning on large-scale image-text datasets to learn powerful vision-language representations, significant improvements in generalization performance have been demonstrated across various downstream tasks. As a result, there has been a growing focus in recent

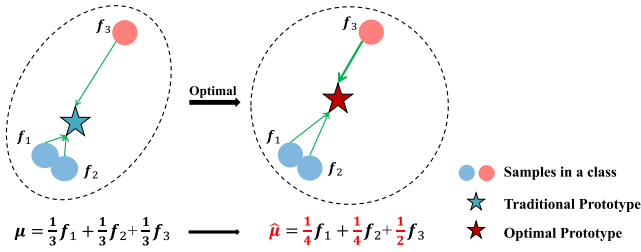


Fig. 2. The calculation of traditional vs optimal prototype.

studies on leveraging pre-trained models to address the challenges associated with long-tailed distribution data. For instance, VL-LTR (Tian, Wang, Zhu, Dai, & Qiao, 2022) introduces a novel vision-language framework based on the foundational model CLIP for long-tail recognition. It initializes with CLIP pre-trained weights, using class-level contrastive learning to connect image and text representations. A language-guided recognition module is then designed, applying attention between image and text features to guide recognition. This establishes a strong benchmark for vision-language-based long-tail visual recognition. LIFT (Shi et al., 2024) suggests that when using foundational model CLIP in long-tail classification, excessive fine-tuning should be avoided, and lightweight fine-tuning strategies should be adopted instead. This helps maintain the model's generalization ability and improves performance on long-tail data.

Almost all of the above SOTA methods rely on reweighting strategies. However, existing reweighting methods fail to accurately estimate the intrinsic space of categories, making it challenging to design appropriate weights. Our proposed method can derive more suitable category weights and, as a plug-and-play solution, can be seamlessly integrated into existing SOTA methods, providing further performance improvements.

3. Analysis of the prototype

For a long-tailed dataset $\{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \{l_i\}_{i=1}^C$, C is the number of classes. Given a sample x_i , we extract its feature $f_i = F(x_i, \theta_f)$, a d -dimensional vector, from the final fully connected layer and calculate the prediction score with $z_i = H(f_i, \theta_h)$, where θ_f and θ_h represent the parameters of the feature extractor and the classifier, respectively. For a class with n samples $\{f_1, f_2, \dots, f_n\}$, its traditional prototype, which indicates the overall diversity, is calculated with:

$$\mu = \frac{1}{n}f_1 + \frac{1}{n}f_2 + \dots + \frac{1}{n}f_n. \quad (1)$$

This prototype is unbiased when the samples are uncorrelated to each other. However, samples in a class are usually correlated due to the human's preference (e.g., front-view angle, and background) when datasets are collected. Therefore, the prototype will be dominated by highly overlapped samples. As shown in Fig. 2, a class contains three samples. When f_1 and f_2 are 90% overlapped but f_3 is very different from $\{f_1, f_2\}$, the traditional prototype is $\mu = \frac{1}{3}f_1 + \frac{1}{3}f_2 + \frac{1}{3}f_3$. In this case, the prototype is more similar to f_1 and f_2 . A more reasonable prototype may be $\hat{\mu} = \frac{1}{4}f_1 + \frac{1}{4}f_2 + \frac{1}{2}f_3$, because it contains almost equal information of these three samples. Therefore, to obtain a more diversified prototype, we should adjust the weights $\{a_i\}_{i=1}^n$ of samples based on their relationships:

$$\hat{\mu} = a_1f_1 + a_2f_2 + \dots + a_nf_n, \quad \sum_{i=1}^n a_i = 1. \quad (2)$$

4. Diversity estimation

In this section, from the perspective of the variance optimization for the prototype, we derive the formula of the *diversity* based on the correlation between samples in a class.

4.1. Stability requirement of the optimal prototype

Suppose infinite instances of a category are collected, then the intrinsic prototype can be obtained, which contains the maximized diversity. In practice, it is impossible to collect unlimited instances, we can only use a limited set at hand to estimate the intrinsic prototype. To approximate the intrinsic prototype, we need to maximize the diversity of the prototype with n collected samples. When the optimal prototype μ^* contains the maximum diversity, it is robust to the collected samples, i.e., no matter how the collected samples of a category vary, it should always be stable.

When the collected samples vary, n samples in the same class can be regarded as n random variables following the distribution of their class. Since f_i is the feature of sample i , we also regard f_i as a d -dimensional random variable. Therefore, we can compute the variance $Var(\hat{\mu})$ to measure the stability of $\hat{\mu}$. The smaller $Var(\hat{\mu})$ is, the more stable $\hat{\mu}$ is. And the more stable $\hat{\mu}$ is, the richer the diversity $\hat{\mu}$ contains. If samples are independent and identically distributed, the optimal μ^* can be calculated with Eq. (1), then, the corresponding variance is:

$$Var(\mu^*) = Var\left(\frac{1}{n}f_1 + \dots + \frac{1}{n}f_n\right) = \frac{1}{n^2} \sum_{i=1}^n \sigma_{f_i}^2 = \frac{\sigma^2}{n}. \quad (3)$$

Since each f_i belongs to the same category (i.e., identically distributed), each σ_{f_i} is equal to σ , where σ is the standard deviation of this category. However, as we have analyzed before, $\{f_1, \dots, f_n\}$ are usually highly correlated in reality. Then, the variance of a weighted prototype $\hat{\mu}$ turns into:

$$\begin{aligned} Var(\hat{\mu}) &= Var(a_1f_1 + a_2f_2 + \dots + a_nf_n) \\ &= \left(\sum_{i=1}^n (a_i)^2 Var(f_i)\right) + \left(\sum_{i,j=1, i \neq j}^n a_i a_j Cov(f_i, f_j)\right) \\ &= \mathbf{a}^T \cdot \mathbb{M} \cdot \mathbf{a}, \end{aligned} \quad (4)$$

where $\mathbb{M}$ is a $n \times n$ covariance matrix and $\mathbf{a} = (a_1, a_2, \dots, a_n)$ is a n -dimensional weight vector. Each diagonal element $\mathbb{M}_{ii} = \sigma^2$. And each non-diagonal element $\mathbb{M}_{ij} = Cov(f_i, f_j)$ is also a scalar. To obtain the optimal prototype μ^* , we should adjust the weights for each sample in Eq. (2) to preserve the maximum diversity. Since μ^* is robust to samples, it can be obtained by minimizing:

$$Var(\mu^*) = \inf \{Var(\hat{\mu})\} = \inf \{\mathbf{a}^T \mathbb{M} \mathbf{a}\}. \quad (5)$$

In Section 4.2, we will discuss how to obtain the $Var(\mu^*)$.

Analysis of the intrinsic prototype. Considering an ideal condition, where infinite samples can be collected for a category. Then, the intrinsic distribution of that category and its corresponding intrinsic prototype can be obtained to reflect the true class distribution. The intrinsic prototype has two key characteristics: (1) Diversity: it captures all potential patterns of feature variations in the category. (2) Stability: it can truly represent the essential characteristics of this class and remains highly robust to sampling bias, even when the collected sample subset is imperfect or skewed.

However, in real-world scenarios, it is impossible to collect infinite samples, and we can only use the limited set at hand to estimate intrinsic prototypes and class distributions. To achieve this goal, this paper optimizes the prototype by weighting the limited sample features to minimize the variance of the optimized prototype. The smaller the variance of a prototype, the more stable it becomes and the closer it approximates the intrinsic prototype, thereby better capturing the essential characteristics of the category.

To intuitively illustrate the differences between prototypes under varying distributions and highlight the motivation and advantages of our ADIR method, we present a comparison of class prototypes under three different sample distributions: the original sample distribution, the weighted-adjusted distribution, and the distribution with sufficient samples under ideal conditions.

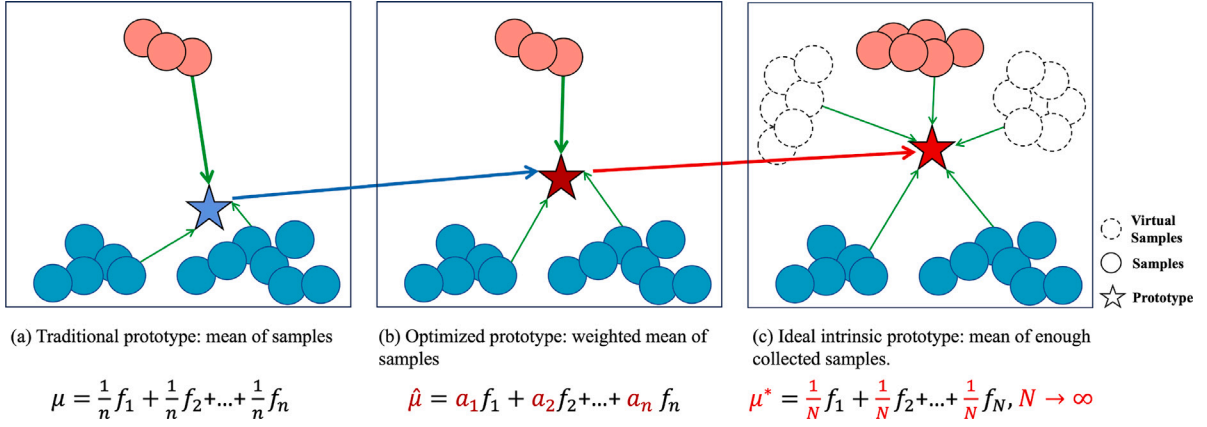


Fig. 3. Visualization of category distribution: (a) Traditional prototype: obtained by directly computing the mean of the observed sample distribution; (b) ADIR-optimized prototype: derived by adjusting the sample distribution using our ADIR method; (c) Ideal intrinsic prototype: derived under the condition of sufficient samples. We observe that the optimized prototype of our ADIR in (b) is closer to the intrinsic class prototype in (c).

(a) Traditional prototype: obtained by directly computing the mean of the observed sample distribution;

(b) ADIR-optimized prototype: derived by adjusting the sample distribution using our ADIR method;

(c) Ideal intrinsic prototype: derived under the condition of sufficient samples.

As shown in Fig. 3, the conventional prototype (a) significantly deviates from the ideal intrinsic prototype (c), while the optimized prototype (b) in ADIR closely approximates the intrinsic prototype (c). This demonstrates that ADIR is effective in reducing the gap between practical and ideal prototype estimation under limited data, which further demonstrates the advantages of our method ADIR.

4.2. Analytical solution to the $Var(\mu^*)$

According to Eq. (4), $\mathbb{M}$ has some easily-proved properties:

(1) $\forall i, j \in [1, n], \mathbb{M}_{ij} \in [-\sigma^2, \sigma^2]$; (2) $\forall i \in [1, n], \mathbb{M}_{ii} = \sigma^2$; (3) $\mathbb{M}^T = \mathbb{M}$; (4) Positive Definite (the proof is in Appendix A.1).

In order to calculate $Var(\mu^*)$ (i.e., $\inf \{a^T \mathbb{M} a\}$), we review the Cauchy-Schwarz inequality. Since $\sum_{i=1}^n a_i = 1$, we can rewrite it as $\mathbf{1}^T a = 1$, where $\mathbf{1} = (1, 1, \dots, 1)^T$ is a n -dimensional vector. According to Cauchy-Schwarz inequality analysis of Eq. (A.2) in Appendix A.2, we can get the following derivation:

$$1^2 = |\mathbf{1}^T a|^2 = |\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}| \underbrace{|\underbrace{a^T \mathbb{M} a}_{Var(\hat{\mu})}}_{(6)}$$

So we can obtain the $\inf \{Var(\hat{\mu})\}$ with Eq. (6):

$$Var(\mu^*) = \inf \{a^T \mathbb{M} a\} = \frac{1}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|}, \quad (7)$$

where $|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}| > 0$, since $\mathbb{M}^{-1}$ is also positive definite.

In the following, we will introduce how to derive *diversity* with the help of $Var(\mu^*)$ without actually calculating μ^* . More explorations about μ^* and a^* will be studied in the future.

4.3. Optimal prototype $Var(\mu^*)$ derived diversity

In order to quantify the *diversity*, we review the Pearson correlation coefficient and use it to reflect the linear correlation of variables f_i and f_j : $\rho_{f_i, f_j} = Cov(f_i, f_j) / \sigma_{f_i} \sigma_{f_j}$. Since f_i and f_j belong to the same class in paper, σ_{f_i} and σ_{f_j} are equal to σ . Then, we can obtain $Cov(f_i, f_j) = \rho_{f_i, f_j} \times \sigma^2$. Therefore, $\mathbb{M}_{ij}$ can be rewritten as:

$$\mathbb{M}_{ij} = \mathbb{P}_{ij} \times \sigma^2, \quad (8)$$

where $\mathbb{P}_{ij}$ represents the correlation of f_i and f_j , and $\mathbb{P}$ denotes the correlation matrix of $\{f_1, f_2, \dots, f_n\}$. Consequently, we can rewrite Eq. (7) as the following:

$$Var(\mu^*) = \frac{1}{|\mathbf{1}^T (\mathbb{P} \times \sigma^2)^{-1} \mathbf{1}|} = \frac{\sigma^2}{|\mathbf{1}^T \mathbb{P}^{-1} \mathbf{1}|}, \quad (9)$$

where $\mathbb{P}^{-1}$ denotes the inverse matrix of $\mathbb{P}$. Suppose the maximum diversity of a category depicted by the n samples is D . Then, we can use D independent and identically distributed virtual samples to represent the maximized spanned space of these n samples. As these D virtual samples are independent and identically distributed, the $Var(\mu^*)$ can be calculated with Eq. (3). Thus, combined with Eq. (9), we can obtain:

$$\frac{\sigma^2}{|\mathbf{1}^T \mathbb{P}^{-1} \mathbf{1}|} = \frac{\sigma^2}{D}. \quad (10)$$

Therefore, the expected *diversity* of a category is:

$$D = |\mathbf{1}^T \mathbb{P}^{-1} \mathbf{1}| = \sum_{i=1}^n \sum_{j=1}^n (\mathbb{P}^{-1})_{ij}. \quad (11)$$

4.4. Correlation coefficient matrix calculation

According to Eq. (11), when the correlation matrix $\mathbb{P}$ of a class is obtained, we can calculate its *diversity*. So we will discuss how to calculate each ρ_{f_i, f_j} of $\mathbb{P}$ in the high-dimensional feature space. Since f_i and f_j from the same class, so $\mu_{f_i} = \mu_{f_j} = \mu$. Then, ρ_{f_i, f_j} can be calculated by:

$$\rho_{f_i, f_j} = \frac{E[(f_i - \mu) \cdot (f_j - \mu)^T]}{\sqrt{E[(f_i - \mu)(f_i - \mu)^T]} \cdot \sqrt{E[(f_j - \mu)(f_j - \mu)^T]}}. \quad (12)$$

In reality, there is only one sample collected for each variable f_i of this class. So we use samples at hand to approximately estimate the ρ_{f_i, f_j} in current datasets as follows:

$$\frac{(f_i - \mu) \cdot (f_j - \mu)^T}{\sqrt{(f_i - \mu)(f_i - \mu)^T} \cdot \sqrt{(f_j - \mu)(f_j - \mu)^T}} = \cos(\theta). \quad (13)$$

Therefore, we can use the cosine similarity of two centralized samples to estimate their correlation.

4.5. Training

After deriving the *diversity* in Eq. (11), we can use it for reweighting. Our ADIR is a two-stage method. In stage-I, we use the original data to train a basic model using the cross-entropy loss. In stage-II, we assign different weights to the loss functions associated with each class. In the following, we will elaborate on the details of the second training phase.

Algorithm 1 Training procedure of ADIR

Input: D_{train} ; $\{(x_i, y_i)\}_{i=1}^N$; ending epochs E_1 and E_2 ; Randomly initialized network parameters θ_f, θ_h ;
Output: Optimized θ_f, θ_h ;

```

1: for  $e = 1 : E_1$  do
2:   for  $b = 1 : B$  do
3:     Sample a batch from  $D_{train}$ ;
4:     Calculate the cross-entropy loss;
5:     Update  $\theta_f, \theta_h$  with SGD;
6:   end for
7: end for
8: for  $e = E_1 : E_2$  do
9:   Calculate diversity with the Algorithm 2;
10:  Calculate weights for each category with Eq. (15);
11:  for  $b = 1 : B$  do
12:    Sample a batch from  $D_{train}$ ;
13:    Calculate weighted loss  $L_{ADIR}$  with Eq. (14);
14:    Update  $\theta_f, \theta_h$  with SGD;
15:  end for
16: end for

```

4.5.1. Adaptive diversity induced reweighting loss

In order to achieve *diversity* balanced training and fairness, the weighted loss function in the second stage can be:

$$\mathcal{L}_{ADIR} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log \frac{e^{z_i^{y_i}}}{\sum_{j=1}^C e^{z_i^j}}, \quad (14)$$

where w_{y_i} is the weight of class l_{y_i} . For the class l_c , the traditional weight based on the *cardinality* is $w_c = \frac{1/n_c}{\sum_{i=1}^C 1/n_i} \times C$, where n_c is the *cardinality*. Differently, we use the *diversity* D_c to replace the *cardinality* to design the weights:

$$w_c = \frac{1/D_c}{\sum_{i=1}^C 1/D_i} \times C. \quad (15)$$

Usually, D_c is calculated by $\sum_{i=1}^n \sum_{j=1}^n (\mathbb{P}_c^{-1})_{ij}$ in Eq. (11), where $\mathbb{P}_c^{-1}$ is the inverse matrix of the $\mathbb{P}_c$. However, estimating $\mathbb{P}_c$ during implementation is challenging. When dealing with large-scale data, it is storage-consuming to maintain high-dimensional features for its numerous samples, particularly for the head class. Additionally, calculating the inverse of a large correlation matrix is both computationally intensive and technically difficult.

4.5.2. Online approximation of the diversity

To address this issue, we simplify the computation by calculating the *diversity* batch by batch according to Eq. (13):

$$D_{cb} = |\mathbf{1}^T \mathbb{P}_{cb}^{-1} \mathbf{1}| = \sum_{i=1}^{n_{cb}} \sum_{j=1}^{n_{cb}} (\mathbb{P}_{cb}^{-1})_{ij}, \quad (16)$$

where n_{cb} is the *cardinality* of class c in the b th batch and $\mathbb{P}_{cb}$ is the correlation matrix of these n_{cb} samples. Then, we sum up D_{cb} in all batches to obtain the whole *diversity* D_c . Finally, we use the obtained D_c of each class for reweighting in the next epoch. Notably, the weights in ADIR can be dynamically adjusted, as the features are refined during training.

4.5.3. Online training algorithm

In summary, the proposed ADIR contains two training stages. In stage-I (lines 1–7 of Algorithm 1), we train the basic classification model using the standard cross-entropy loss. In the second stage (lines 8–16), we use the correlation between samples to calculate the *diversity* of each category and meanwhile use the *diversity*-balanced loss to refine the recognition model. We show the calculation of *diversity* in

Algorithm 2 Calculation of *diversity* in each epoch

Input: D_{train} , and the learned feature extractor θ_f ; B is the number of batches in an epoch;
Output: *Diversity* $\{D_i\}_{i=1}^C$;

```

1: Extract features for all the training samples with  $\theta_f$ ;
2: Calculate the prototype for each category with Eq. (1);
3: for  $b = 1 : B$  do
4:   Sample a batch from  $D_{train}$ , containing a subset of categories  $L_{cb}$ .

5:   for each  $l_c \in L_{cb}$  do
6:     Calculate each term in  $\mathbb{P}_{cb}$  with Eq. (13);
7:     Calculate diversity  $D_{cb}$  with Eq. (16);
8:   end for
9: end for
10: Calculate the diversity for each category with the sum of  $D_{cb}$  in all batches, i.e.,  $D_c = \sum_{b=1}^B D_{cb}$ ;

```

Algorithm 2. According to the Algorithm 1 and 2, we can see that our method relies only on the training dataset, without requiring any additional prior knowledge. Compared to the traditional classification processes, we only need to compute category *diversity* and design loss weights based on the reciprocal of this *diversity*. The *diversity* calculation is based on the features of the training set according to Eqs. (13) and (16).

Note that for a tail class, it is probable that only one instance is sampled in a batch. Then, Eq. (16) degenerates into $D_{cb}=1$. So for most tail classes, *diversity* is similar to *cardinality*. This is reasonable, as head classes are dominated by head classes during the training process, leading to inaccurate features for tail samples and consequently resulting in an inappropriately enlarged *diversity*. In this case, it is reliable to estimate *diversity* for a tail class batch-by-batch. Although *diversity* of the tail is similar to its *cardinality*, the enlarged size of the head depicted by *diversity* can still help to obtain fairer hyperplanes.

Computational complexity analysis. For computational complexity, our model adopts a standard convolutional network architecture, which is commonly used in most existing models, such as Cao et al. (2019), Kang et al. (2020) and Ma, Jiao, Liu, Yang et al. (2023). Our primary contribution is the calculation of category *diversity* based on batch features, without the addition of any new networks. We implement both inference and training by weighting the loss function according to *diversity*.

In the implementation, we compute the *diversity* batch-by-batch according to Eq. (16), where n_{cb} is the *cardinality* of class c in the b th batch and $\mathbb{P}_{cb}$ is the correlation matrix of these n_{cb} samples. And each item ρ_{f_i, f_j} in $\mathbb{P}_{cb}$ can be estimated by cosine similarity of two centralized samples to estimate their correlation. Therefore, our main computational complexity is in the following two aspects:

(1) For each category c in a specific batch, computing an $n_{cb} \times n_{cb}$ matrix $\mathbb{P}_{cb}$ requires $\mathcal{O}(dn_{cb}^2)$ time, where d is the feature dimension.

(2) For each category, computing the inverse of the $n_{cb} \times n_{cb}$ matrix $\mathbb{P}_{cb}$, i.e., $\mathbb{P}_{cb}^{-1}$, requires $\mathcal{O}(n_{cb}^3)$ time.

The overall time complexity of the described algorithm is

$$\mathcal{O}\left(d \sum_{cb} n_{cb}^2 + \sum_{cb} n_{cb}^3\right), \quad (17)$$

where c_b is the number of classes contained in the b -batch. In the worst case, when all samples belong to the same category, the complexity is $\mathcal{O}(b^2 \cdot d + b^3)$. In the average case, when the categories are evenly distributed, the complexity is $\mathcal{O}\left(\frac{b^2 \cdot d}{c_b} + \frac{b^3}{c_b^2}\right)$.

In practice, a batch typically contains multiple categories, with the number of samples n_{cb} per category being relatively small, especially for tail and medium. As a result, the overall complexity remains manageable.

Table 1
Details of CIFAR-LT, ImageNet-LT, iNaturalist 2018.

Datasets	Class number	λ	Train set	Min class size	Max class size	Test set
CIFAR-10-LT	10	200, 100, 50, 20, 10, 1	11,203 ~ 50,000	25 ~ 500	5000	10,000
CIFAR-100-LT	100	200, 100, 50, 20, 10, 1	9502 ~ 50,000	2 ~ 500	500	10,000
ImageNet-LT	1000	256	115,846	5	1280	50,000
iNaturalist 2018	8142	500	437,513	2	1000	24,426

Thus, due to the batch-wise computation, the impact of *diversity* calculation on the overall process is minimal. Consequently, our model exhibits the same algorithmic complexity as these existing models. Furthermore, we can easily combine our *diversity*-based reweighting method to other models, such as ensemble-based NCL (Li, Tan et al., 2022) and CLIP-based method LIFT (Shi et al., 2024), since our main contribution lies in deriving and quantifying class *diversity* based on features, which is calculated in a batch-wise manner.

5. Experiments

5.1. Training details

5.1.1. Datasets

We conduct experiments on four widely used benchmark datasets: CIFAR-10/100-LT (Cui et al., 2019), ImageNet-LT (Liu et al., 2019), and iNaturalist 2018² and two balanced datasets: CIFAR-10/100. We denote the imbalance ratio λ between the sample sizes of the most and least frequent classes as $\lambda = N_{max}/N_{min}$. The overall overview of the dataset is shown in Table 1.

CIFAR-10/100-LT. The original CIFAR-10/100 contains 60,000 images from 10/100 categories with 50,000 for training and 10,000 for validation. The image size is 32×32 . Following (Cui et al., 2019), the long-tailed versions are sampled with a controllable imbalance ratio λ . We can obtain six training sets (Li, Kaixiong et al., 2021) by changing $\lambda \in \{200, 100, 50, 20, 10, 1\}$, where $\lambda = 1$ represents the balanced CIFAR-10/100.

ImageNet-LT. ImageNet is a 1000-class dataset from ILSVRC2012 (Deng et al., 2009), consisting of 1,281,167 training and 50,000 validation images. The original ImageNet is slightly imbalanced with $\lambda = 1.78$. ImageNet-LT is constructed by discarding certain training samples (Liu et al., 2019), resulting in a total of 115,846 images for training, 20,000 for validation, and 50,000 for testing. The largest category contains 1280 samples, while the smallest has just 5 samples, resulting in $\lambda = 256$.

iNaturalist 2018. The iNaturalist 2018 dataset is a large-scale species classification dataset characterized by extreme class imbalance. It contains 435,713 training and 24,424 validation images from 8142 classes with $\lambda = 500$. In our experiments, we use the official training and validation splits.

5.2. Implementation details

We train models on CIFAR-10/100-LT with one NVIDIA 2080Ti GPU, while four NVIDIA 2080Ti GPUs on ImageNet-LT and iNaturalist 2018.

For CIFAR-10/100-LT and balanced CIFAR-10/100, we implement our method with ResNet32. The models are optimized using standard stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay 5×10^{-4} . Following LDAM, the linear warm-up learning rate schedule is adopted (Goyal et al., 2017). Specifically, for CIFAR-10/100-LT, we train the model with a batch size 64 for 200 epochs. The initial learning rate is set to 0.1 which is then decayed by 0.01 at the 160th epoch and the 180th epoch. For balanced CIFAR10/100, following ISDA (Wang et al., 2019), we set the initial learning rate

0.1 and decay it by 0.1 at the 80th epoch and 120th epoch. In order to avoid randomness, we conduct five experiments with five different seeds to train the model and report their average accuracy of them.

For ImageNet-LT, we use ResNet-50 as the backbone and train it 120 epochs. SGD is adopted as the optimizer with the momentum of 0.9 and weight decay of 2×10^{-4} . The initial learning rate is 0.1 and we apply a linear warm-up schedule for the first 5 epochs. Afterward, the learning rate is reduced by a factor of 0.1 at the 60th and 80th epochs.

For iNaturalist 2018, we also utilize ResNet-50 as the backbone, training it with a batch size of 256 using SGD with a momentum of 0.9 and a weight decay of 1×10^{-4} . We employ a linear warm-up learning rate schedule, initializing the learning rate at 0.05.

Resource occupation discussion. Since we calculate the *diversity* of categories based on the batch-wise features of each batch, only a small amount of extra storage is required. Even with large-scale datasets, this does not result in a significant storage burden. In the implementation, a single NVIDIA RTX 3090 GPU was used to process large-scale datasets (i.e., ImageNet-LT and iNaturalist 2018). The diversity computation requires only a small amount of additional storage, such as for storing class prototypes, sample features of the current batch, and the $n_{cb} \times n_{cb}$ correlation matrices of categories in the current batch, where n_{cb} denotes the *cardinality* of class c in the b th batch.

Comparison methods. We compare ADIR with some SOTA reweighting methods. (1) **Baseline:** Cross-Entropy Loss. (2) **Instance-level reweighting** designs different weights on different samples, including Focal loss (Lin et al., 2017), L2RW (Ren et al., 2018), Meta-weight-net (Shu et al., 2019), and FSR (Zhang & Pfister, 2021). (3) **Class-level reweighting** assigns different weights to different classes, including Class-balanced loss (CB) (Cui et al., 2019), LDAM (Cao et al., 2019), Meta-class-weights (Jamal et al., 2020), CE-DSB (Ma, Jiao, Liu, Li et al., 2023) and CE-CR (Ma, Jiao, Liu, Yang et al., 2023). (4) **Hybrid methods** combine the instance-level and class-level methods: LDAM-DRW (Cao et al., 2019), IB (Park et al., 2021), and FSR-DF (Zhang & Pfister, 2021). The methods most closely related to this work are CB and CE-{DRW, CR, DSB}.

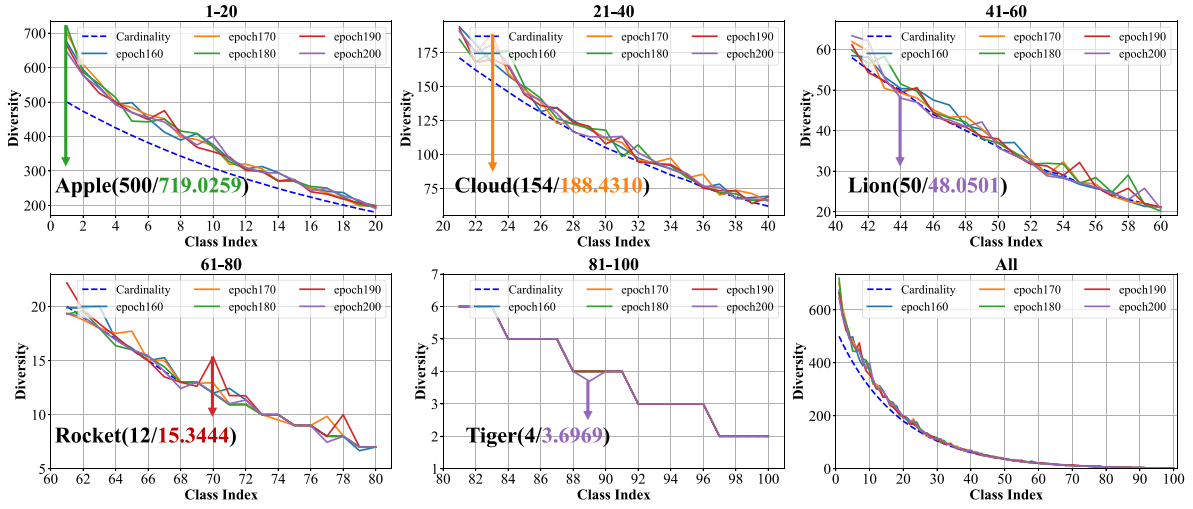
5.3. Results on CIFAR-10/100-LT

We perform comprehensive experiments on CIFAR-LT with $\lambda \in \{200, 100, 50, 20, 10\}$. To avoid randomness, we train the model with five different seeds and report their average accuracy in Table 2. We can draw the following conclusions:

Firstly, our ADIR achieves almost the best results across all the datasets compared with the instance-level reweighting, class-level reweighting, and even the hybrid methods. Compared to the best hybrid method IB, the ADIR improves the accuracy by {1.11%, 1.35%, 1.26%, 1.02%, 0.52%} on CIFAR-10-LT and {6.93%, 6.27%, 6.25%, 4.62%, 3.17%} on CIFAR-100-LT under different values of λ .

Secondly, *diversity* provides a more appropriate measure of a class's spanned size. Compared with the relevant methods CB and CE-{DRW, CR}, ADIR can achieve large improvements. For instance, there are {8.01%, 8.81%, 7.15%, 4.66%, 2.31%}, {-, 6.90%, 7.18%, -, 2.18%} and {7.34%, 7.91%, 7.37%, -, 2.90%} improvements on CIFAR-100-LT, while {6.18%, 5.04%, 3.69%, 2.46%, 1.28%}, {-, 3.27%, 2.99%, -, 1.21%} and {7.17%, 7.01%, 6.76%, -, -0.73%} on CIFAR-10-LT. Besides, we also compare our ADIR with a similar two-stage method Decoupling. Compared to Decoupling, ADIR demonstrates significant performance improvements on both CIFAR-10 and CIFAR-100 datasets,

² https://github.com/visipedia/inat_comp

Fig. 4. Diversity vs cardinality in different epochs on CIFAR-100-LT with $\lambda = 200$.Table 2
Results on CIFAR-10/100-LT.

Imbalanced ratio	CIFAR-10-LT					CIFAR-100-LT				
	200	100	50	20	10	200	100	50	20	10
Cross-Entropy ^b	65.68	70.36	74.81	82.23	86.39	34.84	38.32	43.85	51.14	55.71
Focal ($\gamma = 0.5$) ^b (Lin et al., 2017)	64.00	70.23	76.72	82.89	86.81	35.00	38.69	44.12	51.10	55.70
Focal ($\gamma = 1.0$) ^b (Lin et al., 2017)	65.29	70.38	76.71	82.76	86.66	35.62	38.41	44.32	51.95	55.78
Focal ($\gamma = 2.0$) ^b (Lin et al., 2017)	64.88	69.59	76.52	83.23	86.32	34.75	38.39	43.70	51.02	55.00
L2RW ^c (Ren et al., 2018)	66.25	72.23	76.45	81.35	82.12	33.00	38.90	43.17	50.75	52.12
Meta-weight net ^c (Shu et al., 2019)	67.20	73.57	79.10	84.45	87.55	36.62	41.61	45.66	53.04	58.91
FSR ^a (Zhang & Pfister, 2021)	67.76	–	79.17	–	87.40	35.44	–	42.57	–	55.45
CB ^a (Cui et al., 2019)	68.89	74.57	79.27	84.36	87.49	36.23	39.60	45.32	52.59	57.99
Decoupling (Kang et al., 2020)	66.86	70.62	74.87	81.16	84.18	32.48	34.96	40.22	48.12	52.90
CE-DRW (Cao et al., 2019)	–	76.34	79.97	–	87.56	–	41.51	45.29	–	58.12
CE-DRS (Cao et al., 2019)	–	75.61	79.81	–	87.38	–	41.61	45.48	–	58.11
LDAM ^a (Cao et al., 2019)	–	73.35	–	–	86.96	–	39.60	–	–	56.91
Meta-class-weight ^c (Jamal et al., 2020)	70.66	76.41	80.51	<u>86.46</u>	<u>88.85</u>	<u>39.31</u>	<u>43.35</u>	<u>48.53</u>	<u>55.62</u>	<u>59.58</u>
CE-CR ^a (Ma, Jiao, Liu, Yang et al., 2023)	67.90	72.60	76.20	–	89.50	36.90	40.50	45.10	–	57.40
LDAM-DRW ^a (Cao et al., 2019)	–	77.03	–	–	88.16	–	42.04	–	–	57.99
FSR-DF ^a (Zhang & Pfister, 2021)	66.15	–	79.78	–	88.15	36.74	–	44.43	–	55.60
IB ^a (Park et al., 2021)	<u>73.96</u>	<u>78.26</u>	<u>81.70</u>	85.80	88.25	37.31	42.14	46.22	52.63	57.13
ADIR	75.07	79.61	82.96	86.82	88.77	44.24	48.41	52.47	57.25	60.30

^a Results reported in the original paper.^b Results presented in Cui et al. (2019).^c Results presented in Chen et al. (2022).

with accuracy gains of {8.21%, 8.99%, 8.09%, 5.66%, 4.59%} and {11.76%, 13.45%, 12.25%, 9.13%, 7.40%}, respectively.

Thirdly, ADIR is more effective for handling long-tailed datasets. Compared with CIFAR-10-LT, the improvements on CIFAR-100-LT are more obvious according to the above analysis. Furthermore, ADIR can achieve higher improvements with a larger λ . We believe that this situation occurs because the head with larger *diversity* will severely squeeze the tail. Therefore, *diversity*-balanced reweighting can achieve better improvements.

Overfitting Risk Analysis. We also plot the training and validation loss curves over the entire training process on the CIFAR-100-LT dataset with an imbalance ratio of 100 to analyze the risk of overfitting. As shown in Fig. 5. In the early stages, both training and validation losses decrease steadily, and no signs of overfitting are observed (i.e., the validation loss does not increase while the training loss continues to decrease). Around epoch 160, the validation loss shows a noticeable improvement, which coincides with the introduction of our diversity-based reweighting strategy. After epoch 180, both losses stabilize and converge. This trend provides empirical evidence that our method effectively mitigates the risk of overfitting and promotes better generalization.

Visualizations. To intuitively show that our reweighting method produces more discriminative features, we provide t-SNE visualizations of the feature space on the CIFAR-10 test set with an imbalance ratio of 100 to better illustrate the effectiveness of our re-weighted ADIR method in reducing intra-class variance. As shown in Fig. 6, the features extracted using our ADIR approach form well-separated clusters according to class labels, under class imbalance. This indicates that our method maintains a strong discriminative capability in highly imbalanced scenarios.

5.4. Results on ImageNet-LT and iNaturelist 2018

Table 3 shows the results on ImageNet-LT and iNaturelist 2018. “[a]” and “[b]” denote the results reported in Chen et al. (2022) and original paper, respectively. For ImageNet-LT, we can see that the performance of ADIR boosts to the best 49.48% accuracy. Additionally, ADIR is 8.63% and 2.18% higher than the most relevant method CB and CE-CR, which shows that ADIR can assign more appropriate weights based on the *diversity* of classes for obtaining a fairer classifier. Compared to the method of Decoupling, our approach shows an improvement of 2.18% on ImageNet-LT. For iNaturelist 2018, ADIR

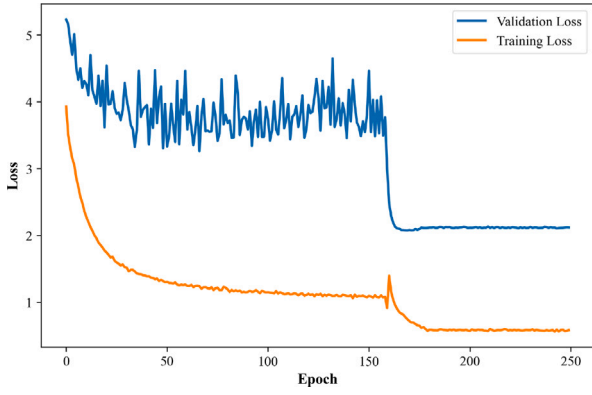


Fig. 5. Training and validation loss curves on the CIFAR-100 dataset with an imbalance ratio of 100.

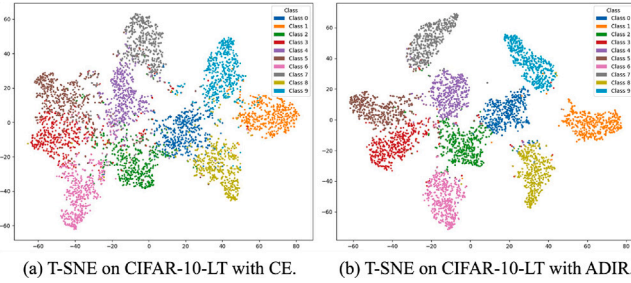


Fig. 6. T-SNE visualizations of the feature space on the CIFAR-10 test set using CE and our ADIR.

Table 3
Results on ImageNet-LT and iNaturalist 2018.

Method	ImageNet-LT	iNaturalist
Cross-Entropy ^b	38.88	57.30
Focal Loss ^b (Lin et al., 2017)	30.50	58.03
CB ^b (Cui et al., 2019)	40.85	61.12
OLTR ^b (Liu et al., 2019)	40.36	—
CE-DRW ^a (Cao et al., 2019)	—	63.73
CE-DRS ^b (Cao et al., 2019)	—	63.56
LDAM ^b (Cao et al., 2019)	41.86	64.58
LDAM-DRW ^a (Cao et al., 2019)	45.74	<u>68.00</u>
Meta-weight ^b (Jamal et al., 2020)	44.92	—
Decoupling ^a (Kang et al., 2020)	47.30	67.60
NCM ^a (Kang et al., 2020)	44.30	63.10
FSR ^a (Zhang & Pfister, 2021)	—	65.52
FSR-DF ^a (Zhang & Pfister, 2021)	—	65.52
IB ^a (Park et al., 2021)	—	65.39
CE-DSB ^a (Ma, Jiao, Liu, Li et al., 2023)	—	64.30
CE-CR ^a (Ma, Jiao, Liu, Yang et al., 2023)	<u>47.30</u>	63.40
ADIR	49.48	68.36

^a Result reported in Chen et al. (2022).

^b Result reported in original paper.

achieves 68.36%, outperforming all other reweighting methods and surpassing the current best hybrid method IB by 3.03%. Moreover, our ADIR surpasses the most relevant CB, CE-DRW, CE-DSB and CE-CR by a large gap of {7.24%, 4.63%, 4.06%, 4.96%}. Additionally, our ADIR outperforms the two-stage methods Decoupling approach by 0.76% on the iNaturalist 2018.

5.5. Difference between diversity and cardinality

To intuitively show the difference between the *diversity* D and the number of samples *cardinality* N , we calculate the *diversity* of each category on CIFAR-100-LT with $\lambda = 200$ under different epochs. As illustrated in Fig. 4, the following observations can be made:

Table 4
Results on balanced CIFAR-10/100.

Method	CIFAR-10	CIFAR-100
Cross-Entropy	92.67	70.53
ADIR	93.03 (+0.36)	71.14 (+0.61)

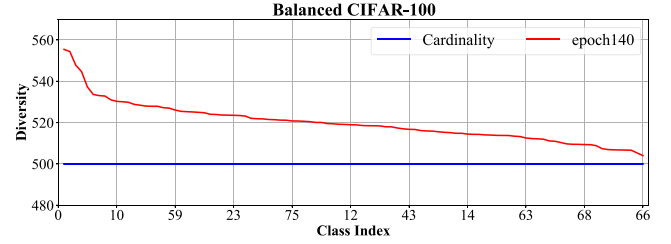


Fig. 7. Diversity vs Cardinality on CIFAR-100.

Firstly, the *diversity* may be larger, smaller, or equal to the N . For example, the N and D of “Apple” are 500 and 719.03 in epoch 180, while they are 154 and 188.4310 for “Cloud” in epoch 170. D is larger than N in both two classes. For “Lion”, $D=48.0501$ is smaller than N because of the limited backgrounds and viewpoints in its samples.

Secondly, the gap between D and N increases for classes with more samples. For instance, the D of “Apple” is 219.03 larger than N , while the D of “Cloud” is 34.431 larger than N . We argue that this is because the “Apple” has sufficient samples ($N = 500$) with richer diversity (such as more backgrounds, colors, etc.) than “Cloud” ($N = 154$).

Thirdly, the rank of several categories in *diversity* is inconsistent with *cardinality*-based rank. For example, the *cardinality*-based rank of “Tiger” is 89, while its *diversity*-based rank becomes 91.

5.6. Results on balanced CIFAR-10/100

We also perform experiments on balanced CIFAR-10/100. Inevitably, even a statistically balanced dataset may exhibit imbalances in *diversity*.

Table 4 shows that ADIR can provide gains of 0.36% and 0.61% over cross-entropy, respectively.

This verifies the effectiveness and flexibility of our ADIR. To clearly show the difference between *diversity* D and *cardinality* N on balanced datasets, we draw them on CIFAR-100 when the epoch = 140. We rerank the classes according to their *diversity*. As shown in Fig. 7, even if the dataset is statistically balanced, it is imbalanced in *diversity*. Moreover, due to the large number of diverse samples collected, D of each category is larger than N .

5.7. Analysis on running time

To clearly demonstrate the running time, we compare the running time of *diversity* in ADIR and curvature in CR (the latest and most effective metric methods) across datasets with varying scales, i.e., CIFAR100 with an imbalanced ratio of 100, the large-scale dataset ImageNet-LT, and iNaturalist 2018. For simplicity, we record the running time (in seconds) for one epoch of ADIR and CR. As shown in Table 5, we can observe that:

For the CIFAR100 dataset, the running time for calculating *diversity* in our ADIR accounts for 4.26% of the total epoch running time, while the curvature computation in CR accounts for 71.93% of the epoch runtime, approximately 16 times greater than our ADIR. For the large-scale ImageNet-LT, the running time spent on curvature calculation in CR constitutes 91.21% of an epoch running time, whereas the diversity calculation in our ADIR occupies only 1.06%. For the even larger iNaturalist 2018 dataset, the diversity calculation in ADIR accounts for just 0.23% while the curvature calculation in CR accounts for 99.99%.

Table 5

Comparison of running time (in seconds) in one epoch on our ADIR and CR with different scale datasets. The proportion represents the proportion of the metric (i.e., *diversity* in ADIR or curvature in CR) calculation in the total epoch running time, i.e., the metric calculation runtime divided by the total epoch time.

Methods	Datasets	Class number	Train set	Metric-calculation (s)	Epoch (s)	Proportion(%)	Results
CR	CIFAR100-LT	100	10,847	25.37	35.27	71.93	40.50
ADIR	CIFAR100-LT	100	10,847	0.59	13.84	4.26	48.41
CR	ImageNet-LT	1000	115,846	1289.25	1413.55	91.21	47.30
ADIR	ImageNet-LT	1000	115,846	1.74	164.85	1.06	49.48
CR	iNaturelist 2018	8142	437,513	448,477,926.93	448,480,655.64	99.99	63.40
ADIR	iNaturelist 2018	8142	437,513	6.14	2633.25	0.23	68.36

Table 6

Combining ADIR with current SOTA methods achieves further improvement.

CIFAR-100-LT	100	50	10
<i>Hybrid rebalancing</i>			
BBN (Zhou et al., 2020)	42.56	47.02	59.12
MiSLAS (Zhong, Cui, Liu, & Jia, 2021)	47.00	52.30	63.20
SAM(Rangwani, Aithal, Mishra, & R., 2022)	44.60	50.00	60.70
SuperDisco (Du, Shen, Zhen, & Snoek, 2023)	50.90	57.20	65.90
CR-CE (Ma, Jiao, Liu, Yang et al., 2023)	40.50	45.10	57.40
<i>Augmentation-based</i>			
FSA (Chu et al., 2020)	48.51	52.17	65.29
RISDA (Chen et al., 2022)	50.16	53.84	62.38
OPeN (Zada, Benou, & Irani, 2022)	51.50	56.30	–
CMO (Park et al., 2022)	47.20	51.70	58.40
GLMC (Du et al., 2023)	57.11	62.32	72.33
<i>Contrastive learning-based</i>			
SSD (Li, Wang, & Wu, 2021)	46.00	50.50	62.30
PaCo (Cui, Zhong, Liu, Yu, & Jia, 2021)	52.00	56.00	64.20
TSC (Li, Cao et al., 2022)	43.80	47.40	59.00
BCL (Zhu, Wang, Chen, Chen, & Jiang, 2022)	51.93	56.59	64.87
<i>Ensemble-based</i>			
RIDE (3 experts) (Wang, Lian, Miao, Liu, & Yu, 2021)	48.60	51.40	59.80
NCL (ensemble) (Li, Tan et al., 2022)	54.20	58.20	65.55
SHIKE(3E) (Jin, Li, Lu, Cheung, & Wang, 2023)	56.30	59.80	–
<i>CLIP-based</i>			
LIFT (Shi et al., 2024)	80.30	82.00	83.80
ADIR	48.41	52.47	60.30
SAM+ADIR	46.20	51.41	61.14
NCL+ADIR	55.23	58.65	65.90
PaCo+ADIR	52.52	56.70	65.47
GLMC +ADIR	58.10	61.58	73.30
LIFT+ADIR	80.80	82.70	84.50

Furthermore, as the dataset scale increases, the proportion of total epoch runtime spent on *diversity* computation in our ADIR remains relatively small. In contrast, the time required for curvature computation in the CR method increases as the dataset size grows. This indicates that, as the dataset scale increases, the batch-based ADIR method is more efficient in computing *diversity* compared to the curvature calculated in the CR method, while only incurring a minimal additional time cost. This further validates the practical efficiency of our ADIR.

5.8. Combining ADIR with SOTA methods

Our ADIR is a simple reweighting method. Compared to the most relevant reweighting methods, ADIR has almost achieved significantly better results (Tables 2–3 in the paper). We agree that our simple and non-parametric reweighting method cannot outperform all the current whistles and bells SOTA methods, especially for the methods with strong data augmentations, multiple models (ensemble methods), contrastive-based methods and CLIP-based methods. Fortunately, our ADIR is a plug-and-play reweighting method that can be flexibly integrated into existing SOTA methods for further enhancement.

To illustrate the flexibility and effectiveness of ADIR, we integrate it with some SOTA methods. Specifically, we combine ADIR with the optimization-based method SAM (Rangwani et al., 2022), the

augmentation-based method GLMC (Du, Yang et al., 2023), the supervised contrastive learning-based PaCo (Cui et al., 2021), the ensemble-based method NCL (Li, Tan et al., 2022), and the CLIP-based approach LIFT (Shi et al., 2024). These methods all leverage *cardinality* for reweighting, whereas we use *diversity* to replace the *cardinality*. As demonstrated in Table 6, our results show that combining ADIR with these SOTA methods leads to consistent performance improvements.

6. Conclusions and future work

In this paper, we have proposed a simple and non-parametric formula to estimate the *diversity* of a class by sufficiently considering the correlations of its samples, which can depict the size of maximized spanned space of a category. By simply assigning the inverse of *diversity* to reweight the loss, our method can outperform existing SOTA reweighting methods on widely used long-tailed datasets. Additionally, it has demonstrated improvements on balanced datasets, suggesting that even balanced datasets may exhibit imbalances in *diversity*.

However, as we have analyzed, the estimated *diversity* for a tail class is suboptimal due to the under-represented tail samples. Therefore, a potential future direction is improving the representation of the tail. Additionally, we will explore more applications about the sample-wise weight a^* and the optimal prototype μ^* . For example, a^* is a good alternative to recognize hard samples. Besides, a^* provides a detailed overview of the samples' distribution of a category, which can instruct the collection of more diversified samples. We believe our ADIR can provide beneficial and interesting insights for researchers in other fields.

CRedit authorship contribution statement

Xiaohua Chen: Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yucan Zhou:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Hongcheng Li:** Visualization, Validation, Methodology, Data curation. **Haihui Fan:** Supervision, Methodology, Formal analysis. **Qinghang Su:** Methodology, Formal analysis, Data curation. **Weiping Wang:** Supervision, Software, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grants 62006221.

Appendix

A.1. Proof of positive definite

In probability theory and mathematical statistics, the covariance matrix is defined as: For any multivariate random d -dimensional n -variable, we can define the $\mathbb{F} = [f_1, f_2, \dots, f_n]$ and $\Omega = [\mu_{f_1}, \mu_{f_2}, \dots, \mu_{f_n}]$, where f_i denotes the single-variable feature, and μ_{f_i} represents the mean when the i -variable takes different values. Both $\mathbb{F}$ and Ω are $n \times d$ matrices. The covariance matrix is then defined as $\mathbb{M} = E[(\mathbb{F} - \Omega)(\mathbb{F} - \Omega)^T]$, which is an $n \times n$ matrix.

Proof. Now, given any non-zero n -dimensional vector α , we have:

$$\begin{aligned} \alpha^T \mathbb{M} \alpha &= \alpha^T E[(\mathbb{F} - \Omega)(\mathbb{F} - \Omega)^T] \alpha \\ &= E[\alpha^T (\mathbb{F} - \Omega)(\mathbb{F} - \Omega)^T \alpha] \\ &= E[(\alpha^T (\mathbb{F} - \Omega))^T ((\mathbb{F} - \Omega)^T \alpha)] \\ &= E[\|(\mathbb{F} - \Omega)^T \alpha\|^2] > 0. \end{aligned} \quad (\text{A.1})$$

In our paper, for a category l_c , $\mathbb{F} = [f_1, f_2, \dots, f_n]$ represents the features of all samples of a class, and $\Omega = [\mu_{f_1}, \mu_{f_2}, \dots, \mu_{f_n}]$ the matrix composed of the corresponding mean value of each random variable. And in our training process, it is impossible for all samples to coincide with the class center. Therefore, in this paper, our covariance matrix $\mathbb{M}$ is **positive definite**, hence Eq. (A.1) must be larger than 0. Besides, since the $\mathbb{M}$ is **positive definite**, the $\mathbb{M}^{-1}$ is also **positive definite**. $\square$

A.2. Cauchy-Schwarz inequality

In order to obtain the optimal prototype, we review the Cauchy-Schwarz inequality. Suppose $\mathbb{Q}$ is a $n \times n$ positive definite matrix, then, we can obtain:

$$\begin{aligned} \alpha^T \mathbb{Q} \beta &= \sum_{i,j=1}^n \mathbb{Q}_{ij} \alpha_i \beta_j \\ &\leq \sqrt{\sum_{i,j=1}^n \mathbb{Q}_{ij} \alpha_i \alpha_j} \sqrt{\sum_{i,j=1}^n \mathbb{Q}_{ij} \beta_i \beta_j} \\ &= \sqrt{\alpha^T \mathbb{Q} \alpha} \sqrt{\beta^T \mathbb{Q} \beta}, \end{aligned} \quad (\text{A.2})$$

where $\alpha, \beta \in \mathcal{R}^{n \times 1}$. Then, with Eq. (A.2), we can obtain that $|\alpha^T \mathbb{Q} \beta|^2 \leq |\alpha^T \mathbb{Q} \alpha| |\beta^T \mathbb{Q} \beta|$. Since $\sum_{i=1}^n \alpha_i = 1$, we can rewrite it as $\mathbf{1}^T \alpha = 1$, where $\mathbf{1} = (1, 1, \dots, 1)^T$ is a n -dimensional vector. According to Eq. (A.2), we can get the following derivation:

$$\begin{aligned} l^2 &= |\mathbf{1}^T \alpha|^2 = \left| \underbrace{\mathbf{1}^T}_{\alpha^T} \underbrace{\mathbb{M}^{-1}}_{\mathbb{Q}} \underbrace{\mathbb{M} \alpha}_{\beta} \right|^2 \\ &\leq \left| \underbrace{\mathbf{1}^T}_{\alpha^T} \underbrace{\mathbb{M}^{-1}}_{\mathbb{Q}} \underbrace{\mathbf{1}}_{\alpha} \right| \left| \underbrace{(\mathbb{M} \alpha)^T}_{\beta^T} \underbrace{\mathbb{M}^{-1}}_{\mathbb{Q}} \underbrace{(\mathbb{M} \alpha)}_{\beta} \right| \\ &= |\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}| |(\alpha^T \mathbb{M}^T)^{-1} (\mathbb{M} \alpha)| \\ &= |\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}| \underbrace{|\alpha^T \mathbb{M} \alpha|}_{\text{Var}(\hat{\mu})}, \end{aligned} \quad (\text{A.3})$$

where the third line of the derivation is based on the Cauchy-Schwarz inequality and $|\cdot|$ stands for determinant. So we can obtain the $\inf \{\text{Var}(\hat{\mu})\}$ with Eq. (A.3):

$$\text{Var}(\mu^*) = \inf \{\alpha^T \mathbb{M} \alpha\} = \frac{1}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|}, \quad (\text{A.4})$$

where $|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}| > 0$, since $\mathbb{M}^{-1}$ is also positive definite. Suppose $\mu^* = \sum_{i=1}^n a_i^* f_i$, then, we have:

$$\begin{aligned} \alpha^{*T} \mathbb{M} \alpha^* &= \inf \{\alpha^T \mathbb{M} \alpha\} = \frac{1}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} = \frac{\mathbf{1}^T \alpha^*}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} \\ &= \frac{\mathbf{1}^T \mathbb{M}^{-1} \mathbb{M} \alpha^*}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} = \frac{(\mathbb{M}^{-1})^T}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} \mathbb{M} \alpha^*. \end{aligned} \quad (\text{A.5})$$

Therefore, when $\alpha^* = \frac{\mathbb{M}^{-1} \mathbf{1}}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|}$, the optimal prototype can be obtained by $\mu^* = \sum_{i=1}^n \frac{(\mathbb{M}^{-1})^T \mathbf{1}}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} f_i$. Note that our original intention is to measure the *diversity* of each category for reweighting. Although we can directly use μ^* for classification, it is too computationally expensive. In the following, we will introduce how to derive *diversity* with the help of $\text{Var}(\mu^*)$ without actually calculating μ^* . More explorations about μ^* and α^* will be studied in the future.

A.3. An example of the calculation of diversity

According to Eq. (11), we can calculate the *diversity* with the correlation matrix. To gain a clearer understanding of the calculation process of diversity, we provided an example.

Example 1. Suppose the correlation matrix of three samples $\{f_1, f_2, f_3\}$ in the same class is:

$$\mathbb{P} = \begin{pmatrix} 1 & 0.2 & 0.4 \\ 0.2 & 1 & 0 \\ 0.4 & 0 & 1 \end{pmatrix}. \quad (\text{A.6})$$

Then, we calculate the $\mathbb{P}^{-1}$, which is the matrix of the inverse of $\mathbb{P}$ as follows:

$$\mathbb{P}^{-1} = \begin{pmatrix} 1.25 & -0.25 & -0.5 \\ -0.25 & 1.05 & 0.1 \\ -0.5 & 0.1 & 1.2 \end{pmatrix}. \quad (\text{A.7})$$

So the *diversity* is $\sum_{i=1}^n \sum_{j=1}^n (\mathbb{P}^{-1})_{ij} = 2.2$. According to Eq. (A.5) in the main body, we can rewrite the optimal α^* as follows:

$$\begin{aligned} \alpha^* &= \frac{\mathbb{M}^{-1} \mathbf{1}}{|\mathbf{1}^T \mathbb{M}^{-1} \mathbf{1}|} = \frac{(\mathbb{P} \times \sigma^2)^{-1} \mathbf{1}}{|\mathbf{1}^T (\mathbb{P} \times \sigma^2)^{-1} \mathbf{1}|} \\ &= \frac{\mathbb{P}^{-1} \mathbf{1}}{\sum_{i=1}^n \sum_{j=1}^n (\mathbb{P}^{-1})_{ij}}. \end{aligned} \quad (\text{A.8})$$

Then, we can obtain that when $\alpha^* = \frac{\mathbb{P}^{-1} \mathbf{1}}{\sum_{i=1}^n \sum_{j=1}^n (\mathbb{P}^{-1})_{ij}} = \{0.5/2.2, 0.9/2.2, 0.8/2.2\} = \{5/22, 9/22, 8/22\}$, the $\text{Var}(\mu^*)$ is attained. So the optimal prototype μ^* is:

$$\mu^* = \frac{5}{22} f_1 + \frac{9}{22} f_2 + \frac{8}{22} f_3. \quad (\text{A.9})$$

Data availability

Data will be made available on request.

References

- Byrd, J., & Lipton, Z. C. (2019). What is the effect of importance weighting in deep learning? In *International conference on machine learning* (pp. 872–881).
- Cao, K., Wei, C., Gaidon, A., Aréchiga, N., & Ma, T. (2019). Learning imbalanced datasets with label-distribution-aware margin loss. In *Advances in neural information processing systems* (pp. 1565–1576).
- Chen, Q., Huang, T., Zhu, G., & Lin, E. (2023). A dual-branch model with inter- and intra-branch contrastive loss for long-tailed recognition. *Neural Networks*, 168, 214–222.
- Chen, X., Zhou, Y., Wu, D., Zhang, W., Zhou, Y., Li, B., et al. (2022). Imagine by reasoning: A reasoning-based implicit semantic data augmentation for long-tailed classification. In *Conference on artificial intelligence* (pp. 356–364).
- Chou, H., Chang, S., Pan, J., Wei, W., & Juan, D. (2020). Remix: Rebalanced mixup. In *European conference on computer vision* (pp. 95–110).
- Chu, P., Bian, X., Liu, S., & Ling, H. (2020). Feature space augmentation for long-tailed data. In *European conference on computer vision* (pp. 694–710).
- Cui, Y., Jia, M., Lin, T.-Y., Song, Y., & Belongie, S. (2019). Class-balanced loss based on effective number of samples. In *IEEE conference on computer vision and pattern recognition* (pp. 9268–9277).

- Cui, J., Zhong, Z., Liu, S., Yu, B., & Jia, J. (2021). Parametric contrastive learning. In *IEEE international conference on computer vision* (pp. 695–704).
- Deng, J., Dong, W., Socher, R., Li, L., Li, K., & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. In *IEEE conference on computer vision and pattern recognition* (pp. 248–255).
- Deng, C., Li, D., Ji, L., Zhang, C., Li, B., Yan, H., et al. (2025). ChatDiff: A ChatGPT-based diffusion model for long-tailed classification. *Neural Networks*, 181, Article 106794.
- Du, Y., Shen, J., Zhen, X., & Snoek, C. G. M. (2023). SuperDisco: Super-class discovery improves visual recognition for the long-tail. In *IEEE conference on computer vision and pattern recognition* (pp. 19944–19954).
- Du, F., Yang, P., Jia, Q., Nan, F., Chen, X., & Yang, Y. (2023). Global and local mixture consistency cumulative learning for long-tailed visual recognitions. In *IEEE conference on computer vision and pattern recognition* (pp. 15814–15823).
- Goyal, P., Dollár, P., Girshick, R., Noordhuis, P., Wesolowski, L., Kyrola, A., et al. (2017). Accurate, large minibatch sgd: Training imagenet in 1 hour. *arXiv preprint arXiv:1706.02677*.
- Huang, C., Li, Y., Loy, C. C., & Tang, X. (2016). Learning deep representation for imbalanced classification. In *IEEE/CVF conference on computer vision and pattern recognition* (pp. 5375–5384).
- Jamal, M. A., Brown, M., Yang, M.-H., Wang, L., & Gong, B. (2020). Rethinking class-balanced methods for long-tailed visual recognition from a domain adaptation perspective. In *IEEE conference on computer vision and pattern recognition* (pp. 7610–7619).
- Jin, Y., Li, M., Lu, Y., Cheung, Y., & Wang, H. (2023). Long-tailed visual recognition via self-heterogeneous integration with knowledge excavation. In *IEEE conference on computer vision and pattern recognition* (pp. 23695–23704).
- Kang, B., Li, Y., Xie, S., Yuan, Z., & Feng, J. (2021). Exploring balanced feature spaces for representation learning. In *International conference on learning representations*.
- Kang, B., Xie, S., Rohrbach, M., Yan, Z., Gordo, A., Feng, J., et al. (2020). Decoupling representation and classifier for long-tailed recognition. In *International conference on learning representations*.
- Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., et al. (2020). Supervised contrastive learning. In *Advances in neural information processing systems*.
- Kim, J., Jeong, J., & Shin, J. (2020). M2m: Imbalanced classification via major-to-minor translation. In *IEEE conference on computer vision and pattern recognition* (pp. 13893–13902).
- Li, T., Cao, P., Yuan, Y., Fan, L., Yang, Y., Feris, R., et al. (2022). Targeted supervised contrastive learning for long-tailed recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 6908–6918).
- Li, M., Cheung, Y., & Hu, Z. (2023). Key point sensitive loss for long-tailed visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4), 4812–4825.
- Li, M., Cheung, Y., & Lu, Y. (2022). Long-tailed visual recognition via Gaussian clouded logit adjustment. In *IEEE conference on computer vision and pattern recognition* (pp. 6919–6928).
- Li, S., Gong, K., Liu, C. H., Wang, Y., Qiao, F., & Cheng, X. (2021). MetaSAug: Meta semantic augmentation for long-tailed visual recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 5212–5221).
- Li, B., Han, Z., Li, H., Fu, H., & Zhang, C. (2022). Trustworthy long-tailed classification. In *IEEE conference on computer vision and pattern recognition* (pp. 6960–6969).
- Li, J., Tan, Z., Wan, J., Lei, Z., & Guo, G. (2022). Nested collaborative learning for long-tailed visual recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 6939–6948).
- Li, T., Wang, L., & Wu, G. (2021). Self supervision to distillation for long-tailed visual recognition. In *IEEE international conference on computer vision* (pp. 610–619).
- Li, B., Yao, Y., Tan, J., Gong, R., Lu, J., & Luo, Y. (2024). Rectify representation bias in vision-language models for long-tailed recognition. *Neural Networks*, 172, Article 106134.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal loss for dense object detection. In *IEEE international conference on computer vision* (pp. 2980–2988).
- Liu, Z., Miao, Z., Zhan, X., Wang, J., Gong, B., & Yu, S. X. (2019). Large-scale long-tailed recognition in an open world. In *IEEE conference on computer vision and pattern recognition* (pp. 2537–2546).
- Ma, Y., Jiao, L., Liu, F., Li, Y., Yang, S., & Liu, X. (2023). Delving into semantic scale imbalance. In *International conference on learning representations*.
- Ma, Y., Jiao, L., Liu, F., Yang, S., Liu, X., & Li, L. (2023). Curvature-balanced feature manifold learning for long-tailed classification. In *IEEE conference on computer vision and pattern recognition*.
- Park, S., Hong, Y., Heo, B., Yun, S., & Choi, J. Y. (2022). The majority can help the minority: Context-rich minority oversampling for long-tailed classification. In *IEEE conference on computer vision and pattern recognition* (pp. 6877–6886).
- Park, S., Lim, J., Jeon, Y., & Choi, J. Y. (2021). Influence-balanced loss for imbalanced visual classification. In *IEEE international conference on computer vision* (pp. 735–744).
- Rangwani, H., Aithal, S. K., Mishra, M., & R., V. B. (2022). Escaping saddle points for effective generalization on class-imbalanced data. In *Advances in neural information processing systems*.
- Ren, J., Yu, C., Sheng, S., Ma, X., Zhao, H., Yi, S., et al. (2020). Balanced meta-softmax for long-tailed visual recognition. In *Advances in neural information processing systems*.
- Ren, M., Zeng, W., Yang, B., & Urtasun, R. (2018). Learning to reweight examples for robust deep learning. In *International conference on machine learning* (pp. 4331–4340).
- Shi, J., Wei, T., Zhou, Z., Shao, J., Han, X., & Li, Y. (2024). Long-tail learning with foundation model: Heavy fine-tuning hurts. In *International conference on machine learning*.
- Shu, J., Xie, Q., Yi, L., Zhao, Q., Zhou, S., Xu, Z., et al. (2019). Meta-weight-net: Learning an explicit mapping for sample weighting. In *Advances in neural information processing systems* (pp. 1917–1928).
- Sinha, S., Ohashi, H., & Nakamura, K. (2022). Class-difficulty based methods for long-tailed visual recognition. *International Journal of Computer Vision*, 130(10), 2517–2531.
- Tian, C., Wang, W., Zhu, X., Dai, J., & Qiao, Y. (2022). VL-LTR: learning class-wise visual-linguistic representation for long-tailed visual recognition. In *European conference on computer vision: Vol. 13685*, (pp. 73–91).
- Wang, Z., Chen, J., & Hoi, S. C. H. (2021). Deep learning for image super-resolution: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(10), 3365–3387.
- Wang, P., Han, K., Wei, X., Zhang, L., & Wang, L. (2021). Contrastive learning based hybrid networks for long-tailed image classification. In *IEEE conference on computer vision and pattern recognition* (pp. 943–952).
- Wang, X., Lian, L., Miao, Z., Liu, Z., & Yu, S. X. (2021). Long-tailed recognition by routing diverse distribution-aware experts. In *International conference on learning representations*.
- Wang, J., Lukasiewicz, T., Hu, X., Cai, J., & Xu, Z. (2021). RSG: a simple but effective module for learning imbalanced datasets. In *IEEE conference on computer vision and pattern recognition* (pp. 3784–3793).
- Wang, Y., Pan, X., Song, S., Zhang, H., Huang, G., & Wu, C. (2019). Implicit semantic data augmentation for deep networks. In *Advances in neural information processing systems* (pp. 12635–12644).
- Wang, Y.-X., Ramanan, D., & Hebert, M. (2017). Learning to model the tail. In *Advances in neural information processing systems* (pp. 7032–7042).
- Wu, T., Huang, Q., Liu, Z., Wang, Y., & Lin, D. (2020). Distribution-balanced loss for multi-label classification in long-tailed datasets. In *European conference on computer vision* (pp. 162–178).
- Wu, Y., Min, F., Zhang, B., & Wang, X. (2023). Long-tailed image recognition through balancing discriminant quality. *Artificial Intelligence Review*, 56(S1), 833–856.
- Xiang, L., Ding, G., & Han, J. (2020). Learning from multiple experts: Self-paced knowledge distillation for long-tailed classification. In *European conference on computer vision: Vol. 12350*, (pp. 247–263).
- Zada, S., Benou, I., & Irani, M. (2022). Pure noise to the rescue of insufficient data: Improving imbalanced classification by training on random noise images. In *International conference on machine learning: Vol. 162*, (pp. 25817–25833).
- Zhang, H., Cissé, M., Dauphin, Y. N., & Lopez-Paz, D. (2018). Mixup: Beyond empirical risk minimization. In *International conference on learning representations*.
- Zhang, Y., Kang, B., Hooi, B., Yan, S., & Feng, J. (2021). Deep long-tailed learning: A survey. *CoRR*, arXiv:2110.04596.
- Zhang, Z., & Pfister, T. (2021). Learning fast sample re-weighting without reward data. In *IEEE international conference on computer vision* (pp. 705–714).
- Zhao, Y., Chen, W., Tan, X., Huang, K., & Zhu, J. (2022). Adaptive logit adjustment loss for long-tailed visual recognition. In *Conference on artificial intelligence* (pp. 3472–3480).
- Zhong, Z., Cui, J., Liu, S., & Jia, J. (2021). Improving calibration for long-tailed recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 16489–16498).
- Zhou, B., Cui, Q., Wei, X., & Chen, Z. (2020). BBN: bilateral-branch network with cumulative learning for long-tailed visual recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 9716–9725).
- Zhu, J., Wang, Z., Chen, J., Chen, Y. P., & Jiang, Y. (2022). Balanced contrastive learning for long-tailed visual recognition. In *IEEE conference on computer vision and pattern recognition* (pp. 6898–6907).